

Final Project Report

Predicting U.S. County-Level Presidential Voting Outcomes
Using Demographic, Socioeconomic, and Historical Election Features

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Dataset	3,113 U.S. counties — ACS 2020 + MIT Election Lab 2000–2024
Training years	2000–2020 (6 election cycles)
Test target	2024 presidential election
Best Classification	XGBoost — Accuracy: 98.6%, F1: 0.952, AUC: 0.998
Best Regression	Gradient Boosting — R ² : 0.991, RMSE: 0.015
Clustering	K-Means k=3 — 99.9% cluster purity
Key finding	dem_share_2020 alone explains 98.9% of 2024 vote share
Novel contribution	edu_x_wins interaction feature ranked #4 in importance

1. Introduction

This report documents the complete machine learning pipeline built for DSC 445 to predict U.S. county-level presidential election outcomes. The project uses ACS 5-Year 2020 Census data combined with MIT Election Lab county-level voting records from 2000 to 2024.

The project is structured in two tiers. The first tier uses only 2020 demographic and socioeconomic features as a baseline. The second tier adds historical election features from 2000–2020 to predict 2024 outcomes — achieving near-perfect prediction accuracy.

1.1 Objectives

- Predict county-level presidential winner (Democrat vs Republican) for 2024
- Predict Democratic vote share percentage for each county in 2024
- Discover natural county groupings through unsupervised clustering
- Quantify the improvement from adding historical voting features over demographics alone
- Identify the most predictive demographic and political trend features

1.2 Data Sources

Source	Description	Years
ACS B01001	Age and sex by county	2020
ACS B02001	Racial composition	2020
ACS B15003	Educational attainment	2020
ACS B19013	Median household income	2020
ACS B17001	Poverty status	2020
ACS B23025	Employment / labor force	2020
ACS B25001	Housing units	2020
MIT Election Lab	County presidential vote counts	2000–2024

2. Data Preprocessing

The preprocessing pipeline loads 8 raw CSV files, cleans all ACS tables through a shared function, fixes two critical bugs from the reference implementation, and produces a single merged county-level dataset.

2.1 Bug Fix 1 — ACS Sentinel Value

The Census Bureau uses -666666666 as a placeholder for suppressed county data. Without replacing this value, all statistical computations on affected columns produce completely wrong results. All occurrences were replaced with NaN before any merging.

2.2 Bug Fix 2 — MIT Election Mode Filter

The reference implementation filtered only mode == TOTAL rows, silently dropping 849 counties that report exclusively by voting mode (absentee, early voting, provisional etc.)

Approach	MIT Counties	Final Merged
Reference (TOTAL only)	2,305	~2,266
Fixed (TOTAL + mode fallback)	3,154	3,115

2.3 Final Dataset

Metric	Value
Total counties	3,115
Total features	162
Missing values	1 (median income, 1 county)
Republican-winning counties	2,576 (82.7%)
Democrat-winning counties	539 (17.3%)
States covered	51 (including DC)

3. Exploratory Data Analysis

3.1 Class Imbalance

Class	Counties	Percentage
Republican (0)	2,576	82.7%
Democrat (1)	539	17.3%
Imbalance ratio	4.78x	—

3.2 Swing County Analysis

Category	Vote Share	Counties	Share
Solid Republican	< 40%	2,204	70.8%
Swing	40%–60%	673	21.6%
Solid Democrat	> 60%	238	7.6%

3.3 Mutual Information Ranking

Rank	Feature	MI Score	Interpretation
1	white_pct	0.1124	Most predictive — negative relationship with Dem vote
2	asian_pct	0.1101	Concentrated in high-Dem urban areas
3	higher_ed_rate	0.1048	Education strongly correlates with Dem voting
4	log_population	0.0868	Larger counties lean Democrat
5	black_pct	0.0730	Higher Black population — more Dem votes
6-9	income, poverty, unemp, density	0.04	Weak standalone predictors

3.4 Key Correlation Findings

- white_pct vs black_pct: -0.81 — avoid using both in linear models
- poverty_rate vs median_income: -0.72 — measure opposite sides of same concept
- white_pct vs dem_vote_share: -0.58 — strongest single linear predictor
- log_population vs dem_vote_share: +0.54 — larger counties lean Democrat

4. Historical Feature Engineering

The key innovation of this project is the construction of a rich historical feature set from MIT Election Lab data across six election cycles (2000–2020). These features capture long-term political trends, short-term momentum, and county stability — far more powerful than demographics alone.

4.1 Per-Year Features

For each election year (2000, 2004, 2008, 2012, 2016, 2020), the following features were extracted:

- `dem_share_{year}` — Democratic vote percentage for that year
- `winner_{year}` — Binary winner (1=Democrat, 0=Republican)

4.2 Engineered Political Trend Features

Feature	Formula	What it captures
<code>dem_wins_count</code>	Sum of <code>winner_{year}</code>	Overall political loyalty
<code>avg_dem_share</code>	Mean of <code>dem_share</code> across years	Long-term political lean
<code>dem_share_trend</code>	<code>dem_share_2020 - dem_share_2000</code>	20-year direction
<code>momentum_2016_2020</code>	<code>dem_share_2020 - dem_share_2016</code>	Recent shift
<code>momentum_2012_2016</code>	<code>dem_share_2016 - dem_share_2012</code>	2012 cycle shift
<code>vote_volatility</code>	Std dev of <code>dem_share</code>	How unpredictable is the county
<code>competitive_count</code>	Elections within 5% of 50%	Genuine competitiveness
<code>political_alignment</code>	Category based on wins	Strong R / Swing / Strong D

4.3 Interaction Features (Novel Contribution)

Four interaction features were engineered that combine political trends with demographic characteristics. These are not present in the reference implementation.

Feature	Formula	Rationale
<code>edu_x_wins</code>	<code>higher_ed_rate x dem_wins_count</code>	Educated AND historically Democrat — strong signal
<code>edu_x_trend</code>	<code>higher_ed_rate x dem_share_trend</code>	Counties getting more educated AND trending Dem
<code>white_x_momentum</code>	<code>white_pct x momentum_2016_2020</code>	White counties that swung in 2020
<code>income_x_volatility</code>	<code>median_income x vote_volatility</code>	Wealthy but unpredictable counties

`edu_x_wins` ranked #4 in Random Forest classification feature importance — ahead of `winner_2016` — confirming that the combination of education level and historical Democratic wins is a highly non-linear

predictor.

4.4 Political Alignment Distribution

Alignment	Counties	Characteristics
Strong Republican	2,078	Won 0/6 elections for Democrat
Lean Republican	403	Won 1-2/6 elections for Democrat
Swing	93	Won 3/6 elections — genuinely competitive
Lean Democrat	231	Won 4/6 elections for Democrat
Strong Democrat	347	Won 5-6/6 elections for Democrat

5. Baseline Models — Demographics Only (2020)

Baseline models were trained using only the 9 engineered demographic features from ACS 2020 data. These establish the performance floor and demonstrate the value added by historical features.

5.1 Baseline Classification

Model	Accuracy	Dem F1	Dem Precision	Dem Recall	ROC AUC
Logistic Regression	0.883	0.727	0.610	0.898	0.952
Random Forest	0.929	0.788	0.820	0.759	0.960
XGBoost	0.917	0.768	0.741	0.796	0.961

5.2 Baseline Regression

Model	R ²	RMSE	MAE
Linear Regression	0.669	0.094	0.074
Ridge	0.669	0.094	0.074
Lasso	0.671	0.094	0.074
Random Forest	0.737	0.084	0.063

5.3 Baseline Clustering

Cluster	Profile	Avg Dem Share	Silhouette
0	Urban/Diverse	0.50	0.309
1	Mixed/Swing	0.42	—
2	Rural/White	0.26	—

6. Historical Models — Predicting 2024

Historical models use the full 32-feature set (19 historical + 9 demographic + 4 interaction) trained on 2000-2020 data to predict 2024 outcomes. All models significantly outperform the demographics-only baseline.

6.1 Historical Classification Results

Model	Accuracy	Dem F1	Dem Precision	Dem Recall	ROC AUC
Logistic Regression	0.979	0.930	0.929	0.931	0.996
Random Forest	0.981	0.935	0.967	0.906	0.998
Gradient Boosting	0.984	0.944	0.966	0.922	0.998
XGBoost	0.986	0.952	0.989	0.917	0.998

6.2 Historical Regression Results

Model	R ²	RMSE	MAE
Linear Regression	0.9909	0.0149	0.0106
Ridge	0.9908	0.0149	0.0106
Lasso	0.9906	0.0151	0.0109
Random Forest	0.9905	0.0152	0.0109
Gradient Boosting	0.9912	0.0146	0.0103
XGBoost	0.9910	0.0148	0.0104

Notably, all 6 regression models achieve $R^2 > 0.99$. Even Linear Regression gets 0.9909 — confirming that dem_share_2020 is so strongly correlated with dem_share_2024 that the relationship is essentially linear.

6.3 Feature Importance Analysis

Rank	Feature	RF Importance	Type
1	dem_share_2020	0.2051	Historical
2	winner_2020	0.1614	Historical
3	dem_share_2016	0.1009	Historical
4	edu_x_wins	0.0991	Interaction (novel)
5	winner_2016	0.0957	Historical

6	avg_dem_share	0.0633	Historical
7	dem_wins_count	0.0500	Historical
8	dem_share_trend	0.0210	Historical
9	edu_x_trend	0.0205	Interaction (novel)
16	higher_ed_rate	0.0084	Demographic

6.4 Historical Clustering

Cluster	Profile	Avg Dem Share 2024	Avg Wins	Political Purity
0	Strong Democrat	0.550	4.5/6	65.8% Democrat
1	Strong Republican	0.248	0.0/6	99.9% Republican
2	Drifting Republican	0.255	0.8/6	99.8% Republican

Cluster 2 is the most politically interesting — these counties have an average `dem_share_trend` of -0.170, meaning they moved heavily Republican over 20 years. These are the classic Obama-to-Trump counties with low education, low income, and high white population percentage.

7. Comprehensive Model Comparison

7.1 Classification — Baseline vs Historical

Model	Demo Only F1	Historical F1	Improvement
Logistic Regression	0.727	0.930	+0.203
Random Forest	0.788	0.935	+0.147
XGBoost	0.768	0.952	+0.184

7.2 Regression — Baseline vs Historical

Model	Demo Only R ²	Historical R ²	Improvement
Linear Regression	0.669	0.991	+0.322
Random Forest	0.737	0.991	+0.254

7.3 Ablation Study

Metric	Full Features	Demo Only	Gain
Classification Accuracy	0.986	0.934	+0.052
Democrat F1	0.952	0.739	+0.213
ROC AUC	0.998	0.960	+0.038
Regression R ²	0.991	0.734	+0.257
Regression RMSE	0.015	0.080	5.3x smaller

7.4 Clustering Comparison

Metric	Demographics Only	Historical Features
Features used	9	26
Silhouette score	0.309	0.183
PCA variance explained	63.5%	58.1%
Cluster 0 purity	Mixed	65.8% Democrat
Cluster 1 purity	Mixed	99.9% Republican
Cluster 2 purity	Mixed	99.8% Republican

Note: The lower silhouette score with historical features is expected — more dimensions make geometric separation harder. However the political purity of clusters is dramatically higher, confirming that historical features produce more meaningful groupings despite the lower silhouette metric.

8. Conclusions and Recommendations

8.1 Key Findings

- Historical voting behavior is by far the strongest predictor — `dem_share_2020` alone has 0.989 importance in regression, explaining nearly all of 2024 vote share variance
- Counties barely change — U.S. county-level political alignment shows strong continuity across election cycles
- `edu_x_wins` interaction feature ranked #4 in classification importance — a novel contribution not present in reference implementations
- Adding historical features improves Democrat F1 by +0.164 and R^2 by +0.254 vs demographics only
- Only 93 genuinely swing counties exist across all 6 election cycles — political polarization is severe
- Cluster 2 identifies drifting Republican counties with avg trend of -0.170 — Obama to Trump voter shift
- All 6 regression models achieve $R^2 > 0.99$ with historical features — even Linear Regression

8.2 Modeling Recommendations

- Use XGBoost with full historical features for classification — 98.6% accuracy, F1 0.952
- Use Gradient Boosting with historical features for regression — R^2 0.991, RMSE 0.015
- Always include `dem_share_2020` — it is the single most powerful predictor
- Include interaction features — `edu_x_wins` and `edu_x_trend` add meaningful non-linear signal
- Evaluate specifically on the 93 swing counties — these are hardest to predict
- Use `class_weight` balanced for all classifiers — 5.89x imbalance in 2024 data
- Do not use `white_pct` and `black_pct` together in linear models — correlation -0.81

8.3 Limitations

- ACS data is from 2020 — demographic features do not reflect 2024 population changes
- 106 counties dropped due to ACS/MIT coverage gaps — primarily Alaska boroughs
- Model relies heavily on past behavior — could fail for genuinely unpredictable election cycles
- County-level analysis masks within-county variation — urban vs suburban areas differ

8.4 Future Work

- Incorporate 2024 ACS estimates when available for more current demographic features
- Add geographic features — proximity to urban centers, state political lean
- Build a county-level prediction map visualization using FIPS codes
- Apply SMOTE oversampling and compare against `class_weight` approach
- Test rolling training pipeline — train on 2000-2016, predict 2020, then retrain for 2024

